In [12]:
```
data = [
    ["ORD101", "2026-10-01", " Laptop ",     55000,    2,    " South ", "Electronic
    ["ORD102", "2026-10-02", "Mouse",          800,    5,    "NORTH",   "Accessorie
    ["ORD103", "2026-10-03", "Keyboard",      1500,    3,    " east ",  "Accessorie
    ["ORD104", "2026-10-04", "Monitor",     12000,    2,    "West",    "Electronics
    ["ORD105", "2026-10-05", "Laptop",       58000,    1,    "south",   "Electronic
    ["ORD106", "2026-10-06", "Headphones",    None,    4,    "North",   "Accessorie
    ["ORD107", "2026-10-07", "Webcam",        2500, None,   "East",    "Accessories
    ["ORD108", "2026-10-08", "Printer",       9000,    2,    None,      "Electronic
    ["ORD109", "2026-10-09", "USB Hub",       1200,    6,    "West",    None],
    ["ORD110", "2026-10-10", "Laptop",      650000,    1,    "South",   "Electronic
    ["ORD111", "2026-10-11", "Mouse",          800,    8,    "NORTH",   "Accessorie
    ["ORD112", "2026-10-12", "Monitor",      12500,    2,    " east ",  "Electronic
    ["ORD113", "2026-10-13", "Keyboard",   "invalid",  4,    "West",    "Accessorie
    ["ORD114", "2026-10-14", "Webcam",         2800,    3,    "South",   "Accessorie
    ["ORD101", "2026-10-01", " Laptop ",     55000,    2,    " South ", "Electronic
]

columns = [
    "order_id", "order_date", "product",
    "price", "quantity", "region", "category"
]

df = pd.DataFrame(data, columns=columns)

df
```

Out[12]:

	order_id	order_date	product	price	quantity	region	category
0	ORD101	2026-10-01	Laptop	55000	2.0	South	Electronics
1	ORD102	2026-10-02	Mouse	800	5.0	NORTH	Accessories
2	ORD103	2026-10-03	Keyboard	1500	3.0	east	Accessories
3	ORD104	2026-10-04	Monitor	12000	2.0	West	Electronics
4	ORD105	2026-10-05	Laptop	58000	1.0	south	Electronics
5	ORD106	2026-10-06	Headphones	None	4.0	North	Accessories
6	ORD107	2026-10-07	Webcam	2500	NaN	East	Accessories
7	ORD108	2026-10-08	Printer	9000	2.0	NaN	Electronics
8	ORD109	2026-10-09	USB Hub	1200	6.0	West	NaN
9	ORD110	2026-10-10	Laptop	650000	1.0	South	Electronics
10	ORD111	2026-10-11	Mouse	800	8.0	NORTH	Accessories
11	ORD112	2026-10-12	Monitor	12500	2.0	east	Electronics
12	ORD113	2026-10-13	Keyboard	invalid	4.0	West	Accessories
13	ORD114	2026-10-14	Webcam	2800	3.0	South	Accessories
14	ORD101	2026-10-01	Laptop	55000	2.0	South	Electronics

In [13]:
```
df.shape
```

Out[13]:
```
(15, 7)
```

Task 1 — Missing Values, Duplicate Orders, Highest Price and Regional Revenue

In [18]:
```
missing_report = pd.DataFrame({
    "Missing_Count": df.isnull().sum()
})

missing_report
```

Out[18]:

	Missing_Count
order_id	0
order_date	0
product	0
price	1
quantity	1
region	1
category	1

Identify duplicate orders

In [19]:
```
df.duplicated().sum()
```

Out[19]:
```
np.int64(1)
```

In [20]:
```
df[df.duplicated(subset="order_id", keep=False)]
```

Out[20]:

	order_id	order_date	product	price	quantity	region	category
0	ORD101	2026-10-01	Laptop	55000	2.0	South	Electronics
14	ORD101	2026-10-01	Laptop	55000	2.0	South	Electronics

Task3 Find the highest product price

In [21]:
```
df["price"] = pd.to_numeric(df["price"], errors="coerce")

highest_price = df["price"].max()

print("Highest product price:", highest_price)
```

```
Highest product price: 650000.0
```

In [22]:
```
df.loc[df["price"].idxmax()]
```

Out[22]:
```
order_id           ORD110
order_date     2026-10-10
product            Laptop
price            650000.0
quantity              1.0
region              South
category      Electronics
Name: 9, dtype: object
```

In [23]:
```
df["quantity"] = pd.to_numeric(df["quantity"], errors="coerce")

df["revenue"] = df["quantity"] * df["price"]
```

```
df[["order_id", "region", "price", "quantity", "revenue"]]
```

Out[23]:

	order_id	region	price	quantity	revenue
0	ORD101	South	55000.0	2.0	110000.0
1	ORD102	NORTH	800.0	5.0	4000.0
2	ORD103	east	1500.0	3.0	4500.0
3	ORD104	West	12000.0	2.0	24000.0
4	ORD105	south	58000.0	1.0	58000.0
5	ORD106	North	NaN	4.0	NaN
6	ORD107	East	2500.0	NaN	NaN
7	ORD108	NaN	9000.0	2.0	18000.0
8	ORD109	West	1200.0	6.0	7200.0
9	ORD110	South	650000.0	1.0	650000.0
10	ORD111	NORTH	800.0	8.0	6400.0
11	ORD112	east	12500.0	2.0	25000.0
12	ORD113	West	NaN	4.0	NaN
13	ORD114	South	2800.0	3.0	8400.0
14	ORD101	South	55000.0	2.0	110000.0

In [24]:

```
region_revenue = (
    df.groupby("region", dropna=False)["revenue"]
      .sum(min_count=1)
      .sort_values(ascending=False)
)

region_revenue
```

Out[24]:

```
region
South      658400.0
 South     220000.0
south       58000.0
West        31200.0
 east       29500.0
NaN         18000.0
NORTH       10400.0
East            NaN
North           NaN
Name: revenue, dtype: float64
```

In [ ]: